THE DUALIZING MODULE REPRESENTS THE FUNCTOR GIVEN BY TOP COHOMOLOGY OF THE FIBRE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AWP (sources/stacks-duality/pinned-original-source.tex line 6157).

Allowed standard prerequisites (editorial).

- Vanishing ranges of the dualizing complex and the interpretation of Ext. Same chapter, Lemma labelled lemma-vanishing-good-dualizing, cited at proof lines 6204-6206, and Derived Categories, Lemma labelled derived-lemma-negative-exts, cited at proof lines 6206-6208. Give the first equality.
- The dualizing complex as the shriek of the base dualizing complex, and adjunction. Same chapter, Lemma labelled lemma-twisted-inverse-image and Section labelled section-duality, cited at proof lines 6210-6214. Give the third and fourth equalities.
- Vanishing of the higher direct images above the fibre dimension. Cohomology of Schemes, Lemma labelled coherent-lemma-higher-direct-images-zero-above-dimension-fibre, cited at proof lines 6214-6218. Gives the final equality.

Source reference. Original section 0AWI: sources/stacks-duality/pinned-original-source.tex, source lines 6070–6070. Same-chapter pointer into the author environment example at source line 6070; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWN: sources/stacks-duality/pinned-original-source.tex, source lines 6125–6125. Same-chapter pointer into the author environment lemma at source line 6125; invoked at proof line 6185 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06XS: sources/stacks-chow/tags.txt, source lines 8861–8861. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6186 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 6190 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0AU3: sources/stacks-duality/pinned-original-source.tex, source lines 5157–5157. Same-chapter pointer into the author environment lemma at source line 5157; invoked at proof line 6191 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02V7: sources/stacks-chow/tags.txt, source lines 3758–3758. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6194 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X/A with dualizing module ω_X be as in Example 0AWI. Let $d = \dim(X_s)$ be the dimension of the closed fibre. If $\dim(X) = d + \dim(A)$, then the dualizing module ω_X represents the functor*

$$\mathcal{F} \longmapsto \mathrm{Hom}_A(H^d(X, \mathcal{F}), \omega_A)$$

on the category of coherent $\mathcal{O}_X$-modules.

Proof. We have

$$\begin{aligned}
\mathrm{Hom}_X(\mathcal{F}, \omega_X) &= \mathrm{Ext}_X^{-\dim(X)}(\mathcal{F}, \omega_X^\bullet) \\
&= \mathrm{Hom}_X(\mathcal{F}[\dim(X)], \omega_X^\bullet) \\
&= \mathrm{Hom}_X(\mathcal{F}[\dim(X)], f^!(\omega_A^\bullet)) \\
&= \mathrm{Hom}_S(Rf_*\mathcal{F}[\dim(X)], \omega_A^\bullet) \\
&= \mathrm{Hom}_A(H^d(X, \mathcal{F}), \omega_A)
\end{aligned}$$

The first equality because $H^i(\omega_X^\bullet) = 0$ for $i < -\dim(X)$, see Lemma 0AWN and Derived Categories, Lemma 06XS. The second equality is follows from the definition of Ext groups. The third equality is our choice of $\omega_X^\bullet$. The fourth equality holds because $f^!$ is the right adjoint of Lemma 0A9E for f, see Section 0AU3. The final equality holds because $R^i f_*\mathcal{F}$ is zero for $i > d$ (Cohomology of Schemes, Lemma 02V7) and $H^j(\omega_A^\bullet)$ is zero for $j < -\dim(A)$. □